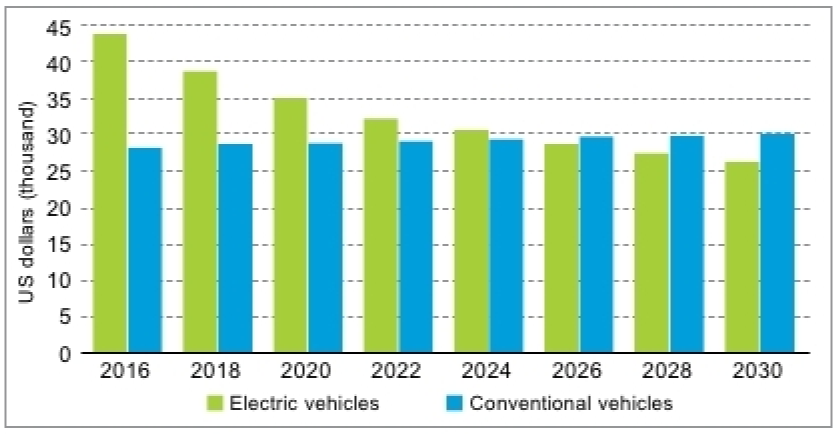

Fig. 3: Four-wheeled electric vehicles to cost less than four-wheeled petrol-fuelled vehicles by 2026 (Source: Bloomberg)

However, a sea of change is anticipated for India's EV industry with a major thrust given by the government.

According to a study by Persistence Research, the Indian EV market is expected to expand at a CAGR of 77 per cent in terms of value during the period 2017-2025. Market players are focusing on expanding their presence in India, where the EV industry is growing at a rapid rate. Key players operating in the market are focussing on entering into tie-ups with local vendors, distributors and aftermarket companies to promote their products.

It is expected to create lucrative opportunities for EV manufacturers as well as vehicle component makers in the near future. Owing to this fact, huge investments are already being planned by many key vendors in the Indian market. The passenger car segment is predicted to be the most attractive segment during the forecast period.

Linchpin policies

On account of pressing environmental concerns, governments across the world are putting regulatory pressures on gasoline vehicle makers, thereby increasing their prices. Governments are also taking initiatives to promote the sale of EVs.

In India too, the government has made a string of announcements that are set to change the landscape of the domestic auto industry. Chief among the announcements are the immediate shift towards Bharat Stage VI emission norms and the planned shift towards only EVs in the longer term. Stricter emission norms, reducing battery prices and increasing consumer awareness are continually driving EV adoption.

The government had initially set a deadline of 2030 by when it planned to remove all fossil fuel-based vehicles from Indian roads. However, a recent announcement has done away with the specific target year; though, it remains aligned with the shift towards EVs, in principle. Individual states have begun to formulate their own EV policies, with Karnataka and Maharashtra leading the pack.

The government of India notified FAME India Scheme [Faster Adoption and Manufacturing of (Hybrid &) Electric Vehicles in India] for implementation with effect from April 1, 2015, with the objective to support hybrid/electric vehicles market development and manufacturing ecosystem. The scheme has four focus areas: technology development, demand creation, pilot projects and charging infrastructure.

Phase I of the scheme was implemented for a period of two years, that is, FY 2015-16 and FY 2016-17 commencing from April 1, 2015. FAME scheme targets the production of ~ 7 million EVs by 2020. Like China, India is also planning to spend largely on subsidising local companies, pushing them to the forefront of electric mobility technologies. FAME India scheme is weighted more towards consumer incentives rather than incentivising local R&D. This makes sense, since the country stands to gain from the technological advances already made globally.

To balance foreign investments and promote domestic industry, the government has also taken several steps, like increasing customs duty on 53 auto components, for complete knock down (CKD) components for cars and commercial vehicles as also on commercial vehicles themselves, providing added protection to the domestic industry and encouraging investors to set up facilities in the country, rather than import goods.

India's EV sales increased 37.5 per cent to 22,000 units during FY 2015-16, and is poised to rise further on the back of cheaper energy storage costs. To keep up with the growing demand, several automakers have started investing heavily in various segments of the industry during the last few months. For example, Mahindra and Mahindra Ltd has partnered with Uber for deploying its electric sedan e-Verito and its hatchback e2o Plus in New Delhi and Hyderabad. The industry attracted foreign direct investments worth US$ 17.91 billion during the period April 2000 - September 2017, according to the data released by Department of Industrial Policy and Promotion (DIPP).

The Centre also boosted demand for EVs by planning to replace its own fleet of vehicles with EVs. Energy Efficiency Services Ltd (EESL), which is a joint venture of public sector undertakings (PSUs) under the power ministry, has floated a global tender for 10,000 EVs to replace the government's fleet.

Challenges to be addressed

Though the automobile landscape is changing, there are still major barriers to the adoption of EVs. Major roadblocks include high prices, inadequate charging infrastructure, limited range that vehicles can run on a single charge and limited available options.

Price. While EVs have a much lower running cost due to cheaper fuel, upfront cost is significantly higher than their petrol counterparts. However, this is set to change as the cost of one expensive component, the battery, is expected to drop over time. Research from Bloomberg New Energy Finance indicates that battery costs (that currently account for about half the cost of EVs) will fall by about 77 per cent between 2016 and 2030, and EVs will cost less than petrol-fuelled vehicles by 2026 (Fig. 3).

A panel headed by cabinet secretary P.K. Sinha has recommended commercial use of ISRO's lithium-ion battery technology under Make In India initiative for EVs, *Livemint* reported. Presently, lithium-ion batteries are not manufactured in India on a commercial basis, and the country has to depend on imports from Japan or China.